Math 240 Matlab Project 2

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*Type/paste Matlab commands for a particular problem/part below the correspondingly labeled line, above the next double % sign. Written responses to questions (not lines of code) should be given as comments, following a single percent symbol % NEWLINE

Problem 1

(a)

```
format short
A = [cos(pi/10) -sin(pi/10); sin(pi/10) cos(pi/10)]
v = [3;5]
A * v
```

```
A =

    0.9511   -0.3090
    0.3090    0.9511


v =

     3
     5


ans =

    1.3081
    5.6823
```

(b)

```
B = [cos(pi/11) -sin(pi/11); sin(pi/11) cos(pi/11)]
AB = A*B
BA = B*A
isequal(AB,BA)
```

```
B =

    0.9595   -0.2817
    0.2817    0.9595


AB =

    0.8255   -0.5644
    0.5644    0.8255


BA =

    0.8255   -0.5644
    0.5644    0.8255


ans =

  logical

   1
```

(c) No, it does not matter since it is commutative as we can see in part b

(d)

```
C = A * B
t = acos(C(1,1))
format rat
t/pi
```

```
C =

    0.8255   -0.5644
    0.5644    0.8255


t =

    0.5998


ans =

      21/110
```

(e)

```
format short
isequal(inv(A), [cos(-pi/10) -sin(-pi/10); sin(-pi/10) cos(-pi/10)])
```

```
ans =

  logical

   0
```

(f)

```
A = [cos(pi/8) -sin(pi/8); sin(pi/8) cos(pi/8)]
L = A * [1 0; 0 -1] * inv(A)


A =

    0.9239   -0.3827
    0.3827    0.9239


L =

    0.7071    0.7071
    0.7071   -0.7071
```

(g)

```
A = [cos(pi/10) -sin(pi/10); sin(pi/10) cos(pi/10)]
AB = A * [cos(0) -sin(0); sin(0) cos(0)]
BA = [cos(0) -sin(0); sin(0) cos(0)] * A
isequal(AB,BA)


A =

    0.9511   -0.3090
    0.3090    0.9511


AB =

    0.9511   -0.3090
    0.3090    0.9511


BA =

    0.9511   -0.3090
    0.3090    0.9511


ans =

  logical
```

```
   1
```

(h)

```
AB
t = acos(C(1,1))
format rat
t/pi
```

```
AB =

    0.9511   -0.3090
    0.3090    0.9511


t =

    0.5998


ans =

      21/110
```

Problem 2

(a)

```
A = [8 4 5;7 6 3;8 6 3]
M = [A eye(3)]
N = rref(M)
X = N(:,4:6)
```

```
A =

       8             4             5
       7             6             3
       8             6             3


M =

  Columns 1 through 5

       8             4             5             1             0
       7             6             3             0             1
       8             6             3             0             0

  Column 6
```

```
       0
       0
       1


N =

  Columns 1 through 5

       1             0             0             0            -1
       0             1             0          -1/6           8/9
       0             0             1           1/3           8/9

  Column 6

       1
    -11/18
    -10/9


X =

       0            -1             1
    -1/6             8/9         -11/18
     1/3             8/9         -10/9
```

(b)

```
inv(A)


ans =

       0            -1             1
    -1/6             8/9         -11/18
     1/3             8/9         -10/9
```

Problem 3

(a)

```
A = [-1 0 0 0;11 3 0 0;-5 6 -2 0;9 2 34 -5]
B = [1 -1 3 2;0 1 -1 2;3 -1 3 1;3 3 -1 3]
det(A)
det(B)


A =

    -1             0             0             0
    11             3             0             0
```

```
      -5           6          -2           0
       9           2          34          -5


B =

       1          -1           3           2
       0           1          -1           2
       3          -1           3           1
       3           3          -1           3


ans =

     -30


ans =

      24
```

(b) Matrix A is triangular so we can just multiply the diagonals

Problem 4

(a)

```
A = [2 3 2 -1;8 3 6 7;2 1 6 8;7 0 5 6]
det(A)
```

```
A =

       2           3           2          -1
       8           3           6           7
       2           1           6           8
       7           0           5           6


ans =

     306
```

(b) i) -306 ii) -612 iii) 306

(c)

```
B = A
B([2 3],:) = B([3 2],:)

C = A
C(3,:) = -2 * C(3,:)
```

```
D = A
D(4,:) = 3 * D(2,:) + D(4,:)
```

```
B =

     2     3     2    -1
     8     3     6     7
     2     1     6     8
     7     0     5     6
```

```
B =

     2     3     2    -1
     2     1     6     8
     8     3     6     7
     7     0     5     6
```

```
C =

     2     3     2    -1
     8     3     6     7
     2     1     6     8
     7     0     5     6
```

```
C =

     2     3     2    -1
     8     3     6     7
    -4    -2   -12   -16
     7     0     5     6
```

```
D =

     2     3     2    -1
     8     3     6     7
     2     1     6     8
     7     0     5     6
```

```
D =

     2     3     2    -1
     8     3     6     7
     2     1     6     8
    31     9    23    27
```

(d)

```
det(B)
det(C)
det(D)
```

```
ans =

    -306
```

```
ans =

    -612
```

```
ans =

     306
```

Problem 5

(a)

```
format short
syms a b c d
A = [a b;c d]
```

```
A =

[a, b]
[c, d]
```

(b)

```
inv(A)
```

```
ans =

[ d/(a*d - b*c), -b/(a*d - b*c)]
[-c/(a*d - b*c),  a/(a*d - b*c)]
```

(c)

```
syms e f g h i
B = [a b c;d e f;g h i]
inv(B)
```

```
B =
```

```
[a, b, c]
[d, e, f]
[g, h, i]


ans =

[ (e*i - f*h)/(a*e*i - a*f*h - b*d*i + b*f*g + c*d*h - c*e*g), -(b*i - c*h)/
(a*e*i - a*f*h - b*d*i + b*f*g + c*d*h - c*e*g),  (b*f - c*e)/(a*e*i - a*f*h
- b*d*i + b*f*g + c*d*h - c*e*g)]
[-(d*i - f*g)/(a*e*i - a*f*h - b*d*i + b*f*g + c*d*h - c*e*g),  (a*i - c*g)/
(a*e*i - a*f*h - b*d*i + b*f*g + c*d*h - c*e*g), -(a*f - c*d)/(a*e*i - a*f*h
- b*d*i + b*f*g + c*d*h - c*e*g)]
[ (d*h - e*g)/(a*e*i - a*f*h - b*d*i + b*f*g + c*d*h - c*e*g), -(a*h - b*g)/
(a*e*i - a*f*h - b*d*i + b*f*g + c*d*h - c*e*g),  (a*e - b*d)/(a*e*i - a*f*h
- b*d*i + b*f*g + c*d*h - c*e*g)]
```

(d)

```
inv(B) * det(B)


ans =

[e*i - f*h, c*h - b*i, b*f - c*e]
[f*g - d*i, a*i - c*g, c*d - a*f]
[d*h - e*g, b*g - a*h, a*e - b*d]
```

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